Type MS · Bubble sort, one pass at a time

Answer 1 of 4 · Question 3 part a.i · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part a.i in this mark scheme extract

3(a)(i)	A1 A2	Award **one** mark for each: • Pass 1 correct (1) • Pass 2 correct (1) 9 11 12 19 14 18 15 17 9 11 12 14 18 15 17 19 (1) 9 11 12 14 15 17 18 19 (1) Must show ONLY the 19 moving in first pass and ONLY the 18 moving in second pass	Ignore lines with individual swaps and only award correct completed passes 9 11 12 19 14 18 15 17 9 11 12 **14** **19** 18 15 17 9 11 12 14 **18** **19** 15 17 9 11 12 14 18 **15** **19** 17 9 11 12 14 18 15 **17** **19** (1) 9 11 12 14 **15** **18** 17 19 9 11 12 14 15 **17** **18** 19 (1) Accept **accurate** right-to-left bubble sort as shown below 9 11 12 19 14 18 15 17 9 11 12 14 19 15 18 17 (1) 9 11 12 14 15 19 17 18 9 11 12 14 15 17 19 18 9 11 12 14 18 15 17 19 (1) Complete sort cannot be the 1st /only line in response – 0 marks as this will be taken as both 18 and 19 moving in first pass	**2**

Type MS · Bubble sort, one pass at a time

Answer 1 of 4 · Question 3 part a.i · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
3(a)(ii)	A3	Award **one** mark for: • 4	If right-to-left pass completed in Q3ai, can only accept 2 swaps	1

Question	mp	Answer	Additional Guidance	Mark
3(a)(iii)	A4	Award **one** mark for: • 3	Do not accept 2 – 3rd pass required for **algorithm** to finish If right-to-left pass completed in Q3ai, can only accept 5 passes	1

Question	mp	Answer	Additional guidance	Mark
3(b)	B1	Award **one** mark for: • Binary search	Accept any other divide and conquer search algorithm (google if necessary) Accept 'binary' on its own as search in question	2
	B2	Award **one** mark for any of the following: • Divide and conquer / list halves each time(1) • Fewer searches needed (1)	Do not accept points about being 'already sorted' as this is in question	

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Type MS · Bubble sort, one pass at a time

Answer 1 of 4 · Question 3 part a.i · 2 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
3(c)		Award **one** mark for each of:		
	C1	(while) Loop ends **only** when 0 entered (1)	Any method e.g. `while age != 0:`	

	C2	Avoiding divide by zero error (1)	By coding 2nd selection box from flowchart – no output required
			Must come after some attempt to process age or count – award even if inside loop.
			ignore 'other' divide by zero errors introduced by candidate
			Must follow logic of flowchart e.g.:
			`if count == 0: (or count != 0)`
			…
			`else`
			…	

	C3	Average value rounded to whole number (1)	Must come after some attempt to process age or count – award even if inside loop. Must target calculated average (even if formula not correct) Any method to get int Accept casting to int *(round not in spec, but string manipulation is)* Accept integer division	**5**

11 of 23

Type MS · Bubble sort, one pass at a time

Answer 1 of 4 · Question 3 part a.i · 2 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part a.i in this mark scheme extract

	C4	Correct value calculated for average and count (1)	Allow even if 0 used to initialise count in B1 (this must be only error in calculation) – set count to -1 to test	
	C5	Count and average included in a syntactically correct final print statement (1)	Award even if never runs Must include both variables (may also include string)	

Correct

```
# Q03c

# program variables
age = 1
count = -1
total = 0

# ------------------------------------------------
# Write your code below this line

while age != 0:

    age = int(input("Please enter age in years: "))

    total = total + age
    count = count + 1

if count == 0:
    print("No ages input")
else:
    average = round(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Typical

```
# Q03c

# program variables
age = 0
count = 0
total = 0

# ------------------------------------------------
# Write your code below this line
age = int(input("Please enter age in years: "))
while age != 0:

    total = total + age
    count = count + 1

    age = int(input("Please enter age in years: "))

if count == 0:
    print("No ages input")
else:
    average = int(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Type MS · Bubble sort, one pass at a time

Answer 1 of 4 · Question 3 part a.i · 2 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part b.i in this mark scheme extract

5 (b)(i)	B1 B2 B3	**Award one mark for each:** • Pass 1 correct in first row of the response (1) • Pass 2 **OR** 3 correct in any row of the response (1) • All passes correct and no extra passes (1) 2 3 8 1 9 10 2 3 1 8 9 10 2 1 3 8 9 10 1 2 3 8 9 10	If the final row (1, 2, 3, 8, 9, 10) is repeated treat as the same row.	(3)

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part b.i in this mark scheme extract

5 (b)(ii)	B4 B5	**Award up to 2 marks for a linked explanation such as:** • The sorting is done in the same place as the original data (1), which means only one additional variable is needed to store the value being swapped (when the values are out of order (1) • All sort and swapping operations are done on the original data (1) which means the amount of memory needed is constant (1) • The list does not need to be split (1), which would use more memory (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part b.i in this mark scheme extract

5 (c)	C1 C2 C3	**Award up to 3 marks for a linked description that includes:** • Start at the beginning of the list (1) • Compare each value in the list with the search item (1) • Repeat until a match is found / the end of the list is reached (1)		(3)

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part b.i in this mark scheme extract

Code - Python

Question	mp	Answer	Additional Guidance	Mark

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part b.i in this mark scheme extract

5 (a)	**Award 1 mark for each of:**			
	A1	Variable to store a number of incorrect passwords (1)		
	A2	Loop used (1)		
	A3	Each password checked to see if the first character is an uppercase letter (1)		
	A4	Each password checked to see if it contains a digit (1)		
	A5	Only check for digits if first character is uppercase OR Only checks if first character is uppercase if at least one digit is present (1)		
	A6	Number of incorrect passwords incremented correctly (1) (allow follow through)		
	A7	All incorrect passwords displayed (1) (allow follow through but must try to display those without uppercase and those without digits)		
	A8	Text file closed (1)		
	A9	Number of incorrect passwords displayed (1) (allow follow through)		(9)
Code examples				
Python				

```
# Add your code here
incorrectPasswords = 0

for password in passwordFile:
    valid = False

    if password[0] in alphabet:
        index = 1
        while (valid == False) and (index in range (1, len(password))):
            if password[index]in digit:
                valid = True
            else:
                index = index + 1

    if not valid:
        incorrectPasswords = incorrectPasswords + 1
        print(password)

print(incorrectPasswords,"passwords are incorrect.")

passwordFile.close()
```

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 6 of 9 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part b.i in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongletters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part b.i in this mark scheme extract

Main program

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

Type MS · Bubble sort, one pass at a time

Answer 2 of 4 · Question 5 part b.i · 3 marks

2022 JUN · Paper 02 · extract 9 of 9 · find part b.i in this mark scheme extract

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Type MS · Bubble sort, one pass at a time

Answer 3 of 4 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part a.ii in this mark scheme extract

<table>
<tr>
<td>3(a)(i)</td>
<td>A1
A2</td>
<td>Award one mark for each:
• Pass 1 correct (1)
• Pass 2 correct (1)
<table>
<tr><td>9</td><td>11</td><td>12</td><td>19</td><td>14</td><td>18</td><td>15</td><td>17</td><td></td></tr>
<tr><td>9</td><td>11</td><td>12</td><td>14</td><td>18</td><td>15</td><td>17</td><td>19</td><td>(1)</td></tr>
<tr><td>9</td><td>11</td><td>12</td><td>14</td><td>15</td><td>17</td><td>18</td><td>19</td><td>(1)</td></tr>
</table>
Must show ONLY the 19 moving in first pass and ONLY the 18 moving in second pass</td>
<td>Ignore lines with individual swaps and only award correct completed passes
<table>
<tr><td>9</td><td>11</td><td>12</td><td>19</td><td>14</td><td>18</td><td>15</td><td>17</td><td></td></tr>
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</table>
Accept accurate right-to-left bubble sort as shown below
<table>
<tr><td>9</td><td>11</td><td>12</td><td>19</td><td>14</td><td>18</td><td>15</td><td>17</td><td></td></tr>
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</table>
Complete sort cannot be the 1st /only line in response – 0 marks as this will be taken as both 18 and 19 moving in first pass</td>
<td>2</td>
</tr>
</table>

9 of 23

Type MS · Bubble sort, one pass at a time

Answer 3 of 4 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part a.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
3(a)(ii)	A3	Award **one** mark for: • 4	If right-to-left pass completed in Q3ai, can only accept 2 swaps	1

Question	mp	Answer	Additional Guidance	Mark
3(a)(iii)	A4	Award **one** mark for: • 3	Do not accept 2 – 3rd pass required for **algorithm** to finish If right-to-left pass completed in Q3ai, can only accept 5 passes	1

Question	mp	Answer	Additional guidance	Mark
3(b)	B1	Award **one** mark for: • Binary search	Accept any other divide and conquer search algorithm (google if necessary) Accept 'binary' on its own as search in question	2
	B2	Award **one** mark for any of the following: • Divide and conquer / list halves each time(1) • Fewer searches needed (1)	Do not accept points about being 'already sorted' as this is in question	

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Type MS · Bubble sort, one pass at a time

Answer 3 of 4 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part a.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
3(c)		Award **one** mark for each of:		
	C1	(while) Loop ends **only** when 0 entered (1)	Any method e.g. `while age != 0:`	

	C2	Avoiding divide by zero error (1)	By coding 2nd selection box from flowchart – no output required
			Must come after some attempt to process age or count – award even if inside loop.
			ignore 'other' divide by zero errors introduced by candidate
			Must follow logic of flowchart e.g.:
			`if count == 0: (or count != 0)`
			…
			`else`
			…	

	C3	Average value rounded to whole number (1)	Must come after some attempt to process age or count – award even if inside loop. Must target calculated average (even if formula not correct) Any method to get int Accept casting to int *(round not in spec, but string manipulation is)* Accept integer division	**5**

11 of 23

Type MS · Bubble sort, one pass at a time

Answer 3 of 4 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part a.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

	C4	Correct value calculated for average and count (1)	Allow even if 0 used to initialise count in B1 (this must be only error in calculation) – set count to -1 to test	
	C5	Count and average included in a syntactically correct final print statement (1)	Award even if never runs Must include both variables (may also include string)	

Correct

```
# Q03c

# program variables
age = 1
count = -1
total = 0

# ------------------------------------------------
# Write your code below this line

while age != 0:

    age = int(input("Please enter age in years: "))

    total = total + age
    count = count + 1

if count == 0:
    print("No ages input")
else:
    average = round(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Typical

```
# Q03c

# program variables
age = 0
count = 0
total = 0

# ------------------------------------------------
# Write your code below this line
age = int(input("Please enter age in years: "))
while age != 0:

    total = total + age
    count = count + 1

    age = int(input("Please enter age in years: "))

if count == 0:
    print("No ages input")
else:
    average = int(total / count)
    print("You entered", count, "ages; the average age is", average)
```

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Type MS · Bubble sort, one pass at a time

Answer 3 of 4 · Question 3 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part a.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Bubble sort, one pass at a time

Answer 4 of 4 · Question 3 part a.iii · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part a.iii in this mark scheme extract

3(a)(i)	A1 A2	Award **one** mark for each: • Pass 1 correct (1) • Pass 2 correct (1) 9 11 12 19 14 18 15 17 9 11 12 14 18 15 17 19 (1) 9 11 12 14 15 17 18 19 (1) Must show ONLY the 19 moving in first pass and ONLY the 18 moving in second pass	Ignore lines with individual swaps and only award correct completed passes 9 11 12 19 14 18 15 17 9 11 12 **14** **19** 18 15 17 9 11 12 14 **18** **19** 15 17 9 11 12 14 18 **15** **19** 17 9 11 12 14 18 15 **17** **19** (1) 9 11 12 14 **15** **18** 17 19 9 11 12 14 15 **17** **18** 19 (1) Accept **accurate** right-to-left bubble sort as shown below 9 11 12 19 14 18 15 17 9 11 12 14 19 15 18 17 (1) 9 11 12 14 15 19 17 18 9 11 12 14 15 17 19 18 9 11 12 14 18 15 17 19 (1) Complete sort cannot be the 1st /only line in response – 0 marks as this will be taken as both 18 and 19 moving in first pass	**2**

Type MS · Bubble sort, one pass at a time

Answer 4 of 4 · Question 3 part a.iii · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part a.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
3(a)(ii)	A3	Award **one** mark for: • 4	If right-to-left pass completed in Q3ai, can only accept 2 swaps	1

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3(a)(iii)	A4	Award **one** mark for: • 3	Do not accept 2 – 3rd pass required for **algorithm** to finish If right-to-left pass completed in Q3ai, can only accept 5 passes	1

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Type MS · Bubble sort, one pass at a time

Answer 4 of 4 · Question 3 part a.iii · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part a.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
3(c)		Award **one** mark for each of:		
	C1	(while) Loop ends **only** when 0 entered (1)	Any method e.g. `while age != 0:`	

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11 of 23

Type MS · Bubble sort, one pass at a time

Answer 4 of 4 · Question 3 part a.iii · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part a.iii in this mark scheme extract

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Correct

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# Q03c

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# Write your code below this line

while age != 0:

    age = int(input("Please enter age in years: "))

    total = total + age
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if count == 0:
    print("No ages input")
else:
    average = round(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Typical

```
# Q03c

# program variables
age = 0
count = 0
total = 0

# ------------------------------------------------
# Write your code below this line
age = int(input("Please enter age in years: "))
while age != 0:

    total = total + age
    count = count + 1

    age = int(input("Please enter age in years: "))

if count == 0:
    print("No ages input")
else:
    average = int(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Type MS · Bubble sort, one pass at a time

Answer 4 of 4 · Question 3 part a.iii · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part a.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark